# Siddhant Agrawal

Computer Science and engineering
Indian Institute Of Technology, Delhi

+91-8284928061
135siddhant@gmail.com
github.com/Siddhant-135


## Education

| Year | Examination/Degree | Institution | Grade |
|---|---|---|---|
| 2024-Present | B.Tech and M.Tech, Computer Science | Indian Institute of Technology, Delhi | 9.89 CGPA |
| 2022-24 | Senior Secondary in CBSE | Bhavan Vidyalaya, Chandigarh | 98 % |
| 2020-22 | Secondary Education in CBSE | Delhi Public School, Chandigarh | 98.4 % |

## Scholastics

- **Department Rank 2** with **CGPA 9.89** in Computer Science. Recipient of IIT Delhi's Merit Award
- Cleared JEE Advanced with **All India Rank 275** amongst 180,200 students.
- **100 percentile with full score in Physics** in JEE Main, and 99.95 percentile overall
- Ranked among **Top 42 Nationally** in **INAO'23** (Penultimate stage of selection for **IOAA** '2023)
- Ranked among **Top 15 Nationally** in **INJSO'21** (Penultimate stage of selection for **IJSO** '2021)
- Team ranked **4th Internationally** at the **Physics Brawl** in 2022 (category C)
- **2X** Qualified **Regional Mathematics Olympiad** (Penultimate stage of selection for **IMO** '2023, 2024)
- **Rank 4** Across IIT Delhi in Simon Marais Mathematics Competition (**SMMC**) 2024-25
- **State Rank 1** in **MIMAMSA'2025**: National Pure Science competition by IISER Pune

## Projects

**AGNI-FIGHTER: Fire Prediction and Modelling** *LSTMs, Cellular Automata, Google Earth Engine*

- Ongoing research project for prediction and modeling of forest fires in Indian Context extending *Our Submission* for ISRO's BAH hackathon using DL and Cellular Automata
- Got up-to-date with current CA research and developed an end-to-end pipeline utilising outputs of a ConvLSTM model with Google Earth Engine and Python to simulate 12-hours of wildfire spread.

**IITD Chatbot and Course Planner** *Langchain, Qdrant, Docker, LiteLLM, CohereEmbed, Regex*

- Co-led development of a **retrieval-augmented generation** based chatbot for **IITD administration** with 0xDevansh delivering command-line prototype with automated mail-scraping and all institute information
- Experimented with various **response aggregation strategies** using Langchain. Currently working to implement ReAct framework and integrate course-scheduling capabilities

**File Management System** *Data Structures, Objected Oriented Programming in C++*

- Developed an in-memory **terminal based version control system** similar to git
- Implemented functions to create files, make changes, view history, snapshot versions and more Github link

**Digital Logic and FPGA System Design** *Verilog HDL, Vivado, Basys3 FPGA*

- Implemented core digital building blocks including logic gates, 7-segment decoders, multi-digit display drivers and debouncers with Verilog. Simulated and Tested everyhting on FPGA Hardware
- Designed and tested Multiply-Accumulate unit (MAC) for Dot-Product operations
- Built memory-driven systems integrating ROM and RAM for vector operations. Github link

**Omnidirectional 3-Wheel Drive Bot** *CAD, Inverse Kinematics, PID control, Simulation*

- Designed 3-wheel omnidirectional drive in FreeCAD, **Simulated** it in **CoppeliaSim** with **terminal-based commands** for testing, and fabricated Arduino-controlled prototype (Robotics Club project)
- Implemented **PID feedback system** for optimal control. Link for Videos and Code

## Relevant Courses

- **Computer Science:** Programming with Python (10/10), Data Structures and Algorithms*, Digital Logic and System Design*, Computer Architecture**, Programming Languages**
- **Mathematics:** Probability and Stochastic Processes*, Discrete Mathematical Structures*, Linear Algebra and Differential Equations(10/10), Introductory Calculus(10/10)
- **Others:** Introduction to Biology*, Intro to Electrical Engineering (10/10) with Lab (10/10), Electrodynamics and Quantum Mechanics (10/10), Engineering Mechanics (10,10)

* refers to ongoing courses, ** refers to courses scheduled for Spring 2026 semester

## Technical Skills

**Programming Languages and Libraries:** Python (PyTorch, Scikit-Learn, Pandas, OpenCV/MediaPipe), C++ , Verilog HDL, Arduino, Scraping (BeautifulSoup, Selenium, Regex), TailwindCSS, Latex

**Software and Tools:** Fusion 360 CAD, FreeCAD, Github, Coppeliasim, AMD Vivado (EDA), Figma